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7 Checklist

1. For all authors...
 - (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]
 - (b) Did you describe the limitations of your work? [Yes] The theoretical results are limited by their assumptions in section 4.
 - (c) Did you discuss any potential negative societal impacts of your work? [Yes] Below the conclusion.
 - (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? [Yes]
2. If you are including theoretical results...
 - (a) Did you state the full set of assumptions of all theoretical results? [Yes] In section 4.
 - (b) Did you include complete proofs of all theoretical results? [Yes] Proofs are given in supplement A.
3. If you ran experiments...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? [Yes] We extended the description of datasets, models and training procedure from section 5 in supplement B and provided the code used in the main paper in the supplemental material.
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? [Yes] Details for the experiments which are not mention in section 5 are given in supplement B.
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? [Yes] Partially, where it was applicable and interesting (c.f. table 2).
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? [Yes] All experiments are run on one NVIDIA GeForce RTX 2080 Ti and package version requirements for the virtual environment are given in the code base.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets...
 - (a) If your work uses existing assets, did you cite the creators? [Yes] Main python packages and models are cited in the paper as well as highlighted in the code.
 - (b) Did you mention the license of the assets? [Yes] The used code bases from others were published under MIT license.
 - (c) Did you include any new assets either in the supplemental material or as a URL? [N/A] Not applicable.
 - (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? [N/A] Not applicable.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [N/A] Not applicable.
5. If you used crowdsourcing or conducted research with human subjects...
 - (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [N/A] Not applicable.
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? [N/A] Not applicable.
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [N/A] Not applicable.

Supplement

A Proofs

In this section we proof the theorems of the main paper and recap the needed definitions to do so.

Definition A.1 (Maximum magnitude attack). Given a multi-class classifier $f(\cdot)$, a loss function $\mathcal{L}(\cdot, \cdot)$, ℓ_p -norm $\|\cdot\|_p$ and a given perturbation strength η the optimization problem for a maximum magnitude attack can be written as

$$\text{maximize } \mathcal{L}(f(x + \delta), y), \text{ w.r.t. } \delta \text{ s.t. } \|\delta\|_p \leq \eta. \quad (5)$$

Theorem A.1 (Sufficient and necessary robustness condition for linear classifiers). *Let $f : \mathbb{R}^d \times \Omega^h \rightarrow \mathbb{R}^k$ be a stochastic classifier with linear discriminant functions and f^A and f^I be two MC estimates of the classifier. Let $x \in \mathbb{R}^d$ be a data point with label $y \in \{1, \dots, k\}$ and $\arg \max_c f_c^A(x) = \arg \max_c f_c^I(x) = y$, and let $x_{adv} = x + \delta^A$ be an adversarial example computed for solving the minimization problem (5) for f^A . It holds that $\arg \max_c f_c^I(x + \delta^A) = y$ if and only if*

$$\min_{c \neq y} \tilde{r}_c^I > \|\delta^A\|_2, \text{ with} \quad (6)$$

$$\tilde{r}_c^I = \begin{cases} \infty, & \text{if } \cos(\alpha_c^{I,A}) = \frac{\langle -\nabla_x(f_y^I(x) - f_c^I(x)), \delta^A \rangle}{\|\nabla_x(f_y^I(x) - f_c^I(x))\|_2 \cdot \|\delta^A\|_2} \leq 0 \\ \frac{f_y^I(x) - f_c^I(x)}{\|\nabla_x(f_y^I(x) - f_c^I(x))\|_2 \cdot \cos(\alpha_c^{I,A})}, & \text{otherwise,} \end{cases}$$

where $\alpha_c^{I,A}$ is the angle between $-\nabla_x(f_y^I(x) - f_c^I(x))$ and δ^A .

Proof. An adversarial attack on f^I with the adversarial example $x + \delta^A$ is not successful iff $\forall c \in \{1, 2, \dots, k\}, c \neq y$:

$$f_y^I(x + \delta^A) - f_c^I(x + \delta^A) > 0.$$

With Taylor expansion around x we can rewrite $f_y^I(x + \delta^A) - f_c^I(x + \delta^A)$ as

$$\begin{aligned} & f_y^I(x) + \langle \nabla_x f_y^I(x), \delta^A \rangle - f_c^I(x) - \langle \nabla_x f_c^I(x), \delta^A \rangle \\ &= f_y^I(x) - f_c^I(x) + \langle \nabla_x f_y^I(x) - \nabla_x f_c^I(x), \delta^A \rangle \\ &= f_y^I(x) - f_c^I(x) - \|\nabla_x(f_y^I(x) - f_c^I(x))\|_2 \cdot \|\delta^A\|_2 \cdot \cos(\alpha_c^{I,A}) \end{aligned} \quad (7)$$

where $\alpha_c^{I,A} := \angle(-\nabla_x(f_y^I(x) - f_c^I(x)), \delta^A)$. We can distinguish two cases for each c .

Case 1: $\cos(\alpha_c^{I,A}) \leq 0$. In this case last term of eq. (7) is negative or zero and thus

$$f_y^I(x + \delta^A) - f_c^I(x + \delta^A) \geq f_y^I(x) - f_c^I(x) > 0,$$

where the second inequality holds since $\arg \max_c f_c^I(x) = y$.

Case 2: $\cos(\alpha_c^{I,A}) > 0$. In this case the last term of equation eq. (7) is positive and thus

$$f_y^I(x + \delta^A) - f_c^I(x + \delta^A) \leq f_y^I(x) - f_c^I(x).$$

As we see from rearranging eq. (7), it holds $f_y^I(x + \delta^A) - f_c^I(x + \delta^A) > 0$ if

$$\tilde{r}_c^I := \frac{f_y^I(x) - f_c^I(x)}{\|\nabla_x(f_y^I(x) - f_c^I(x))\|_2 \cdot \cos(\alpha_c^{I,A})} > \|\delta^A\|_2. \quad (8)$$

For each class c either case 1 holds and we define $\tilde{r}_c^I := \infty$, or condition (8) is fulfilled, which yields the condition stated in the theorem. $\square$

In the main part of the paper we relaxed the linear classifier assumption by L -smoothness, which was defined as follows: